

Harsha kumar

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Senior Software Engineer with 12 years of experience building scalable distributed systems, high-performance APIs, cloud-native applications using .NET, Angular and Python. Experienced in microservices, Azure/AWS, performance engineering, and resilient system design, with a strong focus on AI applications using LLMs, RAG and vector databases.

Technical Skills

Languages : C#, Python, JavaScript, TypeScript, T-SQL

Backend : .NET 6/8, ASP.NET Core, Entity Framework Core, ADO.NET, LINQ, Dependency Injection, RESTful APIs, Azure API Management

Frontend : Angular, Vue.js, Ionic, HTML, CSS, Bootstrap

AI & LLMs : Microsoft Foundry (Azure AI Foundry), Microsoft Agent Framework, Azure OpenAI, OpenAI API, Claude API, LangChain, Semantic Kernel, Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Workflows, Prompt Engineering, Vector Search

Database & Caching : SQL Server, PostgreSQL, MongoDB, Redis, Elasticsearch

Cloud Platforms : Azure (App Service, Functions, API Management, AKS, Key Vault, Application Insights), AWS (EC2, Lambda, API Gateway, S3, RDS, SQS, SNS)

Containers & DevOps : Docker, Kubernetes, Helm, Jenkins, GitHub Actions, CI/CD pipelines

Messaging & Event Streaming : MSMQ, Apache Kafka, Azure Service Bus, AWS SQS, AWS SNS

Architecture : System Design, Microservices, Clean Architecture, CQRS, Distributed Systems, Event-Driven Architecture

Engineering : DSA, Design Patterns, SOLID, Async & Parallel Programming

Reliability & Observability : Resiliency, Fault Tolerance, AppDynamics, OpenTelemetry, Grafana, Application Insights, Performance Engineering

AI-Assisted Development Tools : GitHub Copilot, Claude Code, ChatGPT/Codex

Experience & Projects

Diaspark - Verizon (client)

Apr 2022 – Present

Senior Software Engineer

Verizon Home App | C# .Net 6/8, Angular 18, MongoDB, SQL Server, AWS, Azure, Jenkins, Bitbucket, 42Crunch

- Led backend development for scalable, secure, high-performance .NET microservices, mentoring engineers, driving architectural decisions, conducting code reviews, aligning technical solutions with product and reliability goals.
- Designed and delivered RESTful APIs for real-time router management, supporting router status, connected devices, parental controls, Wi-Fi settings, QoS, and Internet Backup services.
- Integrated Bitdefender, BCS, and other third-party REST APIs to extend platform capabilities for network security, device protection, and router management while maintaining scalable and maintainable service integrations.
- Implemented fault-tolerant resilience strategies using Polly and .NET 8 Resilience Engine, including retry, circuit breaker, timeout, and fallback policies, reducing production service disruptions by 50%.
- Served as the go-to engineer for complex .NET issues, guiding architectural decisions and conducting code reviews to ensure scalability, performance, and secure coding standards across the application lifecycle.
- Developed a high-performance wrapper library to interface with BCS, improving API response time by 25%.
- Architected and deployed cloud-native APIs on Azure using App Service and API Management, integrating Key Vault for secure secret management and Application Insights for monitoring, diagnostics, and observability, improving system reliability and operational visibility.
- Designed and implemented distributed caching mechanism using in-memory and Redis cache, reducing database load by 60% and slashing API response time by 40% under peak load conditions.

Wells Fargo International Solutions Pvt Ltd

Sep 2021 – Apr 2022

Senior Software Engineer

ACM - Ad Campaign Management | C# .Net 5, Angular 12, SQL Server, Azure, SSIS, SonarQube

- Developed and maintained full-stack modules for an internal Ad Campaign Manager platform using .NET Core API and Angular enabling end-to-end management of digital ad campaigns.
- Built campaign creation and scheduling workflows with support for ad targeting, frequency capping and budget constraints, improving marketing efficiency.
- Leveraged Azure Blob Storage for asset uploads (creatives, banners), ensured secure file access using SAS tokens.
- Optimized campaign data retrieval using LINQ and EF Core with server-side pagination and filtering, enhancing load times by 40%.

Eurofins IT Solutions India Pvt Ltd

Feb 2020 – Sep 2021

Technology Specialist - Performance Engineering Practices

Key Highlights

- Recommended and implemented .NET performance optimization techniques such as connection pooling, caching strategies, async programming and object reuse.
- Identified and mitigated memory leaks, CPU spikes and thread contention issues through advanced diagnostics and code instrumentation.
- Delivered detailed RCA reports and performance tuning recommendations to product owners and leadership.
- Conducted end-to-end performance assessment using tools like AppDynamics, dotTrace and SQL Profiler leading to up to 50% improvement in response times.
- Took initiative to create an SEBP handbook covering coding standards and performance tips—leading to a 30% reduction in post-deployment defects and improved consistency, quality and scalability.

Bill Scanner | Ionic, Angular 9, SQLite, HTML, CSS

- Led the design and development of a cross-platform mobile application for digitizing and managing bill receipts.
- Gathered requirements, performed technical analysis and defined the app's architecture using Ionic and Angular.
- Integrated local storage with SQLite for offline capability and seamless data access.
- Integrated native device capabilities (camera, file access) using Capacitor plugins for scanning and uploading bills.
- Set up CI/CD pipeline for automated build and deployment using GitHub Actions and Ionic Appflow.
- Deployed the application successfully to both Apple App Store and Google Play Store, adhering to compliance and review standards.  Bill Scanner (iOS)  Bill Scanner (Android)

Ernst Young Global Delivery Services LLP

Aug 2018 – Jan 2020

Senior Software Engineer - Tax Transformation Technology

Work Flow Automation | C# .Net Core 2.1, Angular 6, SQL Server, Azure WebApp, WebJobs, AppInsights

- Designed and implemented a workflow automation platform using .NET Core, EF Core, Angular and SQL Server to streamline internal business processes across audit and tax teams.
- Developed a reusable library leveraging Microsoft Exchange Web Services (EWS) to perform read, write, and delete operations on shared mailboxes, supporting automation of mailbox management tasks.
- Developed a data visualization dashboard using ng2-charts, providing stakeholders with real-time insights and improving decision-making processes.
- Developed Escalation and ticket generation algorithm.

Arcadix Infotech Pvt Ltd

Jan 2017 – Aug 2018

Software Engineer

Test Drive - School Mgmt, Assessment Platform | C# .Net Core 1.1, Angular 2, SQL Server, HTML, CSS, Bootstrap

- Developed a browser-based school and assessment platform using .NET Core, Angular and SQL Server covering daily operations across students, staff, administration and entire test life cycle.
- Built a Learning Content Management System (LCMS) with a WYSIWYG editor to enable test creators to build assessments with minimal effort.
- Optimized SQL queries and used EF Core LINQ projections to improve load time for dashboards and reports with large datasets.

Intellileap Solutions India Pvt Ltd

Dec 2014 – Dec 2016

Software Developer

iLeap - BPMS | C# .Net framework, ASP.NET, Web API, CanJs, WinForms, SQL Server, HTML, CSS, Bootstrap

- Developed and enhanced a custom Business Process Management System using .Net API, CanJs and SQL Server enabling enterprise clients to model, automate and monitor complex workflows.
- Built dynamic workflow modules allowing users to design and configure custom business processes using a drag-and-drop UI and rule-based logic engine.
- Developed a WinForms-based Import-Export tool to accelerate implementation and deployment processes, while enhancing debugging efficiency for automated workflows.
- Developed API integrations with third-party systems CRM plugins, PDF viewers and DocuWare DMS as part of automated processes.

Education

Visvesvaraya Technological University

July 2014

Bachelor of Engineering in Computer Science

- **Coursework:** OOPs, Data Structures, Algorithms, DBMS, Web development, System Design, OS